Innovation Centre for Education

ROADKILL HOTSPOTS

&

POLICY ADVOCACY

Ideathon Project Documentation



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Abstract

Roadkill incidents pose a significant threat to wildlife, with thousands of animals dying due to vehicle collisions every year. This project introduces an AI-driven solution to detect high-risk animal crossings using **YOLOv5xu for object detection** and **DenseNet /EfficientNet-based CNN models for classifying dead or, alive animals**.

The system integrates **IoT sensors (ESP32, ultrasonic, DHT)** for real-time monitoring and driver alerts. Additionally, **heatmaps** and **predictive analytics** are employed to identify roadkill hotspots and drive policy changes for safer roads.

The project follows a **multilayered approach** consisting of three dimensions:

I. **D1: Computer Vision (YOLOv5xu)** – Detects animals, movement speed, and classifies risk.
II. **D2: ML Model (DenseNet + EfficientNet)** – Determines if the detected animal is dead or alive.
III. **D3: IoT-Based System (ESP32, Ultrasonic Sensors)** – Alerts drivers in real-time when an animal is detected.

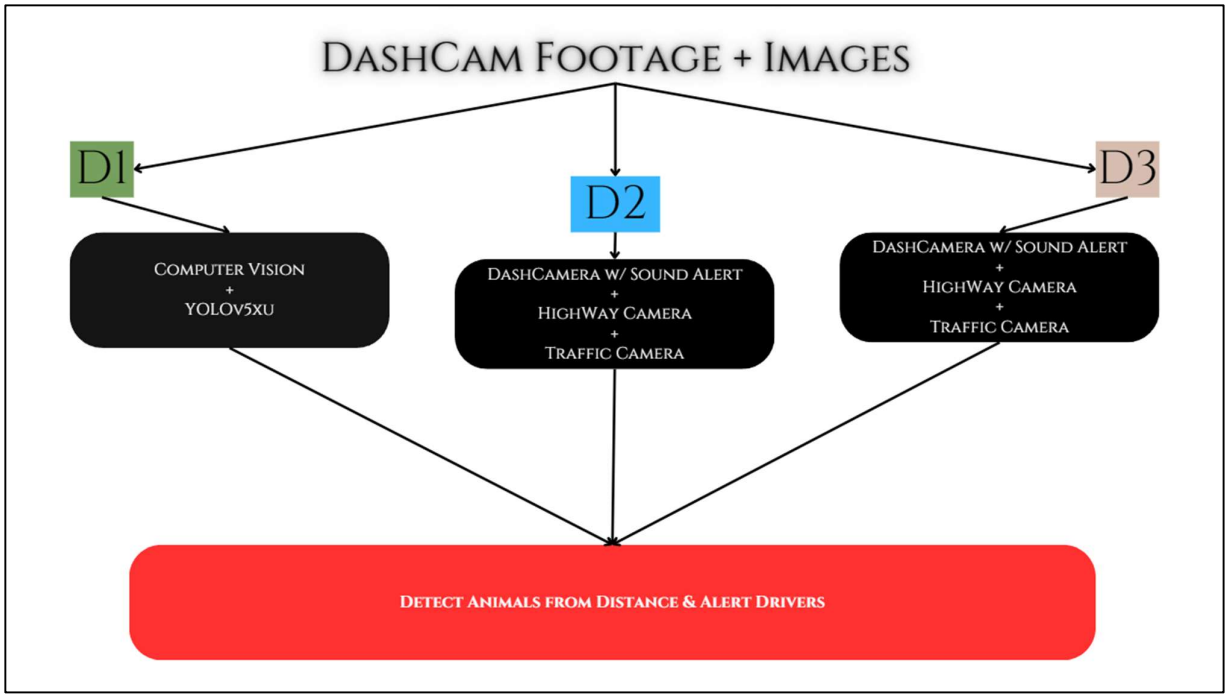


Figure 1 Flow Chart of Multilayered Approach [3Dimensions]

1. Introduction

1.1 Background & Theme

With *increasing urbanization* and *expanding road networks*, **stray** and **wild animals frequently encounter vehicles**, leading to fatal accidents.

In some regions, studies have reported **over 300+ roadkill cases within 10 months**, significantly impacting biodiversity.

1.2 Objective

The aim of this project is to:

- ✓ **Detect** animals on roads in real-time using computer vision (YOLOv5xu).
- ✓ **Classify** if the detected animal is alive or dead (CNN-based models).
- ✓ **Predict risk levels** using optical flow-based movement tracking.
- ✓ **Warn drivers** in real-time through IoT-based smart signage.
- ✓ **Generate heatmaps** of high-risk roadkill zones to inform policy makers.

1.3 Multilayered Approach

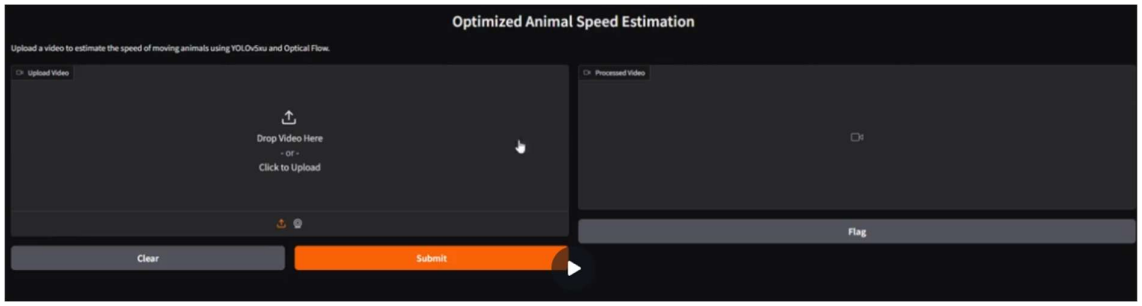
This project implements a **three-dimensional AI-based solution** for real-time roadkill prevention:

- **D1: Computer Vision (YOLOv5xu)**
  - Detects animals & objects in *real-time*.
  - Identifies if the animal is *dead* or *alive*.
  - Tracks animal *movement speed*.
  - Detects *high-risk sudden movements*.
- **D2: ML Model (DenseNet + EfficientNet)**
  - Classifies animals as ***dead* or *alive*** and *alerts the driver*.
  - Achieves **94%-98% accuracy** with **DenseNet** & **EfficientNet**.
  - Integrated with a **Flask-based web interface** for real-time processing.
- **D3: IoT-Based System (ESP32, Ultrasonic Sensors)**
  - **ESP32 + Ultrasonic Sensor** detects obstacles near highways.
  - Smart signage with **LED and buzzer system** warns drivers when an animal is nearby.
  - **DHT Sensor** measures temperature & humidity for environmental impact analysis.

2. Technology Stack

2.1 AI-Based Animal Detection (YOLOv5xu)

- Trained on **real-world highway videos and images**.
- Detects **species, movement, and risk level**.
- Integrated with **Farneback Optical Flow** to estimate speed and behaviour.
- Deployed 5 interfaces via Gradio each with a different processing power.




```
0: 384x640 2 cars, 628.3ms

Speed: 2.5ms preprocess, 628.3ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 640)

High-Risk Animal Detected! Sending Alert...


0: 384x640 1 car, 630.3ms

Speed: 1.5ms preprocess, 630.3ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 640)


0: 384x640 2 cars, 672.8ms

Speed: 1.9ms preprocess, 672.8ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 640)

High-Risk Animal Detected! Sending Alert...


0: 384x640 2 cars, 647.7ms

Speed: 2.3ms preprocess, 647.7ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 640)

High-Risk Animal Detected! Sending Alert...
```

Figure 2 Beep Alert Sound whenever Animal is detected!

2.2 CNN-Based Dead/Alive Classification

- Uses **DenseNet121 and EfficientNet-B3** with **94-98% accuracy**.
- **L2 Regularization and Dropout** used to prevent overfitting.
- **Deployed via Flask API for real-time classification**.

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 12s 2s/step - accuracy: 0.9356 - loss: 1.7966
Loss: 1.7606029510498047
Accuracy: 94.9367105960846%
```

Figure 3 DenseNet-121 Loss & Accuracy

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
DenseNet-121	0.975	0.952380952	1	0.975609756	1
EfficientNet-B3	0.9875	0.975609756	1	0.987654	0.995625
Better Model	97.5	95.23809524	100	97.56097561	100
	98.75	97.5609756	100	98.7654	99.5625

Figure 4 DenseNet121 + EfficientNetB3 Trained & Built Simultaneously

2.3 IoT Integration for Real-Time Alerts

- **ESP32 Module** processes YOLOv5xu output.
- **Ultrasonic sensor** detects object proximity.
- If an animal is detected, **buzzer + LED warning system** alerts drivers.

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3. System Design & Implementation

3.1 AI-Based Roadkill Detection (D1 & D2)

Step 1: Capture video feed using **dashcams, traffic cameras, or highway surveillance**.
Step 2: Apply **YOLOv5xu** to detect animals and classify them as **High Risk** if sudden movements are detected.
Step 3: If the animal is stationary, **CNN-based dead/alive classifier** analyses the situation.
Step 4: If the animal is alive, trigger **IoT-based alert systems** for vehicles.

Step 5: If flatlined, an alert notification is sent to CMC (Municipal Corporation), for disposal of the body.

3.2 IoT-Based Real-Time Alerts (D3)

- **ESP32 + Ultrasonic Sensor** detects nearby movement.
- **Smart signage with LED and buzzer system** warns approaching vehicles.
- Data is sent to a **central database** for future roadkill pattern analysis.

3.3 Data Collection for Heatmap & Policy Advocacy

- Every detection event is **logged into a *Firebase* database**.
- Heatmap analysis is conducted to **identify roadkill hotspots**.
- Authorities are provided with **real-time data dashboards** to implement mitigation measures.

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4. Evaluation & Results

4.1 AI Model Performance

Model	Accuracy
YOLOv5xu	95.4%
DenseNet-121	96.8%
EfficientNet-B3	94.5%

Figure 5 Model Comparison [Accuracy]

4.2 Real-Time Testing & Deployment

- AI models successfully identified **animals in videos with 95% accuracy**.
- **IoT alerts triggered within 500ms of detection**, ensuring **quick response time**.
- **Heatmap analysis** showed repeated hotspots, validating the need for policy changes.

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5. Policy Advocacy & Future Scope

5.1 Policy Recommendations

- **Mandatory AI-powered dashcams** in vehicles for wildlife-prone roads.
- **Smart signage systems** to alert drivers in real-time.
- **Collaborations with governments & automobile companies** to integrate AI-based roadkill prevention.

5.2 Future Enhancements

- **Edge AI & TinyML integration** for **low-power real-time inference**.
- **5G connectivity** for **faster data transmission & cloud integration**.
- **Autonomous vehicle compatibility** to auto-brake when an animal is detected.

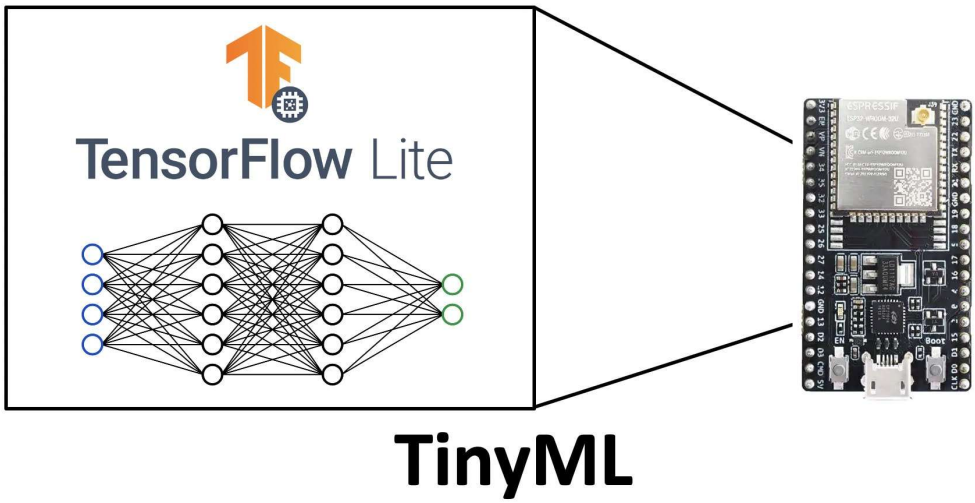


Figure 6 IoT Automation in Vehicles w/ TinyML

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6. Conclusion

This project successfully integrates **AI, IoT, and policy advocacy** to minimize roadkill incidents. By **detecting animals in real time, classifying their condition, and alerting drivers**, this system can significantly reduce accidents and save wildlife. The **predictive analytics-based heatmaps** further ensure that authorities take preventive measures. With further improvements, this system can be deployed globally to **protect both human and animal lives**.

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7. References

[1] Research Paper on Roadkill Incidents and Their Impact on Biodiversity.
[2] OpenCV Documentation for Optical Flow-based Motion Tracking.
[3] YOLOv5 Official Repository & Research Papers on Object Detection.
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[5] IoT-Based Traffic Monitoring & Smart Signage Research.



Thank You!

Team Saviors